RAMA SAI ANUDEEP ITHA

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PROFESSIONAL SUMMARY

Software Engineer with 6+ years building highly scalable distributed systems and full-stack applications serving millions of users. Proven track record delivering exceptional results: achieved 95% operational efficiency improvement, reduced deployment cycles by 50%, and maintained 99.99% uptime for infrastructure processing 10M+ events/second. Deep technical expertise in Python, Java, React, distributed systems architecture, and cloud infrastructure (AWS) with strong focus on system reliability, performance optimization, and data-driven decision making. Experienced leading cross-functional projects, mentoring engineers, and driving measurable business impact through engineering excellence. Passionate about solving complex technical challenges at scale and building products that create meaningful impact for billions of users.

PROFESSIONAL EXPERIENCE

Software Development Engineer | Amazon Web Services (CloudFront), Seattle, WA | *Jun 2023 - Present*

- Developed CMAP, a centralized PaaS for CloudFront that automated global resource planning using Java, React, and AWS, leveraging TDD to achieve 90% unit/integration test coverage and boosting operational efficiency by 95%
- Managed end-to-end ML pipelines for global capacity forecasting, processing large-scale historical and real-time telemetry data using Python (scikit-learn, pandas, NumPy), optimizing resource allocation and accelerating deployment velocity by 50%
- Built highly scalable distributed systems processing 10M+ events/second across 450+ global edge locations, designing microservices architecture with focus on reliability, performance, and fault tolerance achieving 99.99% uptime
- Led design reviews for key platform features, demonstrating strong technical judgment, system design capabilities, and ability to navigate complex architectural trade-offs in distributed systems
- Ensured resilience through game-day testing, stress testing, and chaos engineering for streaming events including Super Bowl and Thursday Night Football, maintaining system reliability during peak demand with millions of concurrent users
- Collaborated cross-functionally with product managers, engineering teams, and operations to define requirements, align on technical solutions, and deliver high-impact features on aggressive timelines
- Mentored junior engineers on distributed systems design, code quality, testing practices, and performance optimization, conducting thorough code reviews and fostering engineering excellence

Lead Software Engineer (Founder) | RoomieHub, Seattle, WA | *Apr 2024 - Jul 2025*

- Architected and led development of cross-platform mobile application (React Native) and cloud-based distributed backend (Node.js, AWS) serving 5,000+ users with 99.99% system reliability
- Designed scalable backend microservices using Node.js, PostgreSQL, MongoDB, and AWS (Lambda, API Gateway, RDS), achieving <200ms response times and seamless horizontal scalability
- Implemented OAuth 2.0 and JWT authentication ensuring data privacy and security, built real-time features using WebSocket protocols, and deployed infrastructure using Docker and AWS
- Demonstrated end-to-end ownership: defined product requirements, designed system architecture, implemented features, established CI/CD pipelines, and iterated based on user feedback and data analytics

Software Engineer II | JP Morgan Chase & Co., Bangalore, India | *Jul 2019 - Aug 2021*

- Engineered automated pipeline in Java and Spring Boot reducing deployment cycles from days to minutes, decreasing manual effort by 90% while maintaining operational security through ADFS and OAuth 2.0 migrations
- Led implementation of high-throughput, fault-tolerant Infrastructure as a Service (IaaS) platform processing thousands of transactions per second with 99.99% uptime across global infrastructure
- Built distributed microservices using Java/Spring Boot with relational databases (MySQL, PostgreSQL), implementing complex SQL queries and database optimizations improving performance by 90%
- Collaborated with distributed teams across US, India, and UK on architectural decisions, code reviews, and technical implementations, demonstrating strong cross-functional communication and leadership

Machine Learning Engineer Intern | Simbo.ai, Bangalore, India | *May 2019 - Jul 2019*

- Built and deployed production deep learning model using CNNs, Python, and TensorFlow for speech recognition supporting 15 Indian languages, collaborating on model evaluation and performance optimization workflows

Full-Stack Engineer | Inspiredge IT Solutions, Visakhapatnam, India | *May 2017 - May 2018*

- Architected and engineered 20+ high-performance full-stack websites using React.js, AngularJS, Django, and MySQL, delivering mobile-responsive experiences and scalable backend solutions for global clients

KEY TECHNICAL PROJECTS

Centralized Event Management System | Sep 2023 - Dec 2023

- Designed and implemented high-availability event management system using Java and serverless/containerized architectures (AWS Lambda, ECS, EC2) for automated global capacity planning across CloudFront infrastructure
- Built secure, resilient infrastructure leveraging AWS CDK, SQS, and SNS for event orchestration with real-time monitoring via CloudWatch, ensuring 99.99% system reliability and availability

Automated Server Planning & Deployment System | Jan 2024 - Jun 2024

- Engineered automated deployment framework integrating capacity forecasts with real-time server metrics to dynamically scale infrastructure across global regions, ensuring continuous service availability
- Designed proactive capacity provisioning system leveraging event-driven demand modeling, accelerating rollout velocity by 50% while maintaining optimal buffer capacity for peak traffic events

Automated CPU Efficiency System | Jul 2024 - Dec 2024

- Built automated system calculating per-server CPU efficiency by processing three months of high-velocity metrics (event load, request volume, byte-throughput) aggregated at minute-level granularity
- Optimized global traffic distribution by integrating predictive efficiency data into routing layer, enabling balanced load distribution across cities, countries, and regions while maintaining ultra-low latency

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, Go, C++, SQL, Kotlin
Frontend & Mobile: React.js, React Native, AngularJS, TypeScript, HTML5, CSS3
Backend & Frameworks: Spring Boot, Node.js, Express, Django, Flask, Microservices, RESTful APIs, GraphQL, gRPC
Distributed Systems: Scalable architecture, High availability, Fault tolerance, Load balancing, Consensus protocols, Event-driven systems, Message queues (Kafka, SQS, SNS)
Databases: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Cassandra, SQL optimization, NoSQL, Database design
Cloud & DevOps: AWS (Lambda, ECS, EC2, S3, RDS, DynamoDB, CloudWatch, CDK), Docker, Kubernetes, CI/CD, Infrastructure as Code, Linux
ML & Data: TensorFlow, scikit-learn, pandas, NumPy, ML pipelines, Data processing, ETL, Analytics
Testing & Quality: TDD, Unit testing, Integration testing, JUnit, Selenium, Jest, Load testing, Stress testing, Chaos engineering
Tools & Practices: Git, GitHub, Jira, Agile/Scrum, Code reviews, Design reviews, Technical documentation, Monitoring & alerting

KEY ACHIEVEMENTS & IMPACT

- 95% Operational Efficiency Improvement: Transformed manual capacity planning into automated system serving 500+ engineers globally
- 50% Deployment Velocity Increase: Accelerated server rollout across global infrastructure through ML-driven forecasting and automation
- 99.99% System Uptime: Maintained exceptional reliability during peak events (Super Bowl, TNF) with millions of concurrent users
- 90% Manual Effort Reduction: Automated enterprise deployment pipelines reducing cycles from days to minutes at JP Morgan
- 10M+ Events/Second at Scale: Built distributed systems processing high-velocity telemetry across 450+ global edge locations
- 5,000+ Users Served: Founded and scaled RoomieHub from zero to production with complete end-to-end ownership

PROFESSIONAL CERTIFICATIONS

AWS Certified Solutions Architect - Associate | AWS Certified AI Practitioner

EDUCATION

Arizona State University, Tempe, Arizona | Master of Science in Computer Science | Aug 2021 - May 2023
VNIT Nagpur, Maharashtra, India | Bachelor of Technology in Computer Science | Jul 2015 - May 2019